Report 1: Thematic Analysis — Research Motivation Categories

Analyst: AI (MiniMax M2.7, via Zo Computer)

Source: 27 raw OCR text files from `Documents/OCR 1.1/Raw/`

Date: 2026-04-28

Section A: Category Counts and Poster Lists

Category | Count | Posters |

|—|—|—|

Basic Biology | 13 | Berbouti et al. (2025) - CCC FA25, Brown et al. (2025) - CCC SP25, Camara et al. (2025) - LU SP25, Dehuelbes et al. (2025) - LU SP5, Diaz et al. (2025) - CCC SP25, Ford et al. (2023) - CCC SP23, Godoy-Pena et al. (2025) - CCC FA25, Henriquez et al. (2025) - COD WI24, Logan et al. (2025) - CCC FA25, Luera et al. (2025) - CCC FA25, Nii et al. (2025) - CCC FA25, Otala et al. (2025) - LU SP25, Sakana et al. (2025) - CCC SP25 |

Neurological & Psychiatric Disease | 6 | Avetisyan et al. (2023) - CCC SP23, Cabral et al. (2025) - CCC FA25, Gill & Alcazar (2025) - CCC FA25, Haubelt & Alcazar et al. (2025) - CCC FA25, Holmes (2025) - CCC SP25, Tuttle et al. (2025) - CCC SP25 |

Cancer | 2 | Meraz et al. (2023) - CCC SP23, Paramo-Ojeda et al. (2025) - CCC SP25 |

Metabolic & Digestive Disease | 3 | Lemus et al. (2025) - FA25 CCC, Pedireddi et al. (2025) - CCC SP25, Rodriguez (2025) - CCC SP25 |

Inherited / Genetic Disorder | 3 | Paderna et al. (2025) - CCC SP25, Trevino et al. (2022) - CCC SP22, Zlaket et al. (2023) - CCC SP23 |

Other | 0 | |

Total posters analyzed: 27

Section B: Category Definitions

Basic Biology

Research focused on understanding fundamental biological mechanisms — such as gene expression patterns, regional tissue specialization, signaling pathways, or developmental processes — without explicitly targeting a specific human disease. These posters use model organisms to explore how genes and cells normally function. Example topics: regional gut gene expression profiling, immune pathway profiling, neuropeptide signaling biology, paralog/ortholog characterization, or pigment/melanin biology without explicit disease framing.

Neurological & Psychiatric Disease

Research motivated by understanding the molecular basis of brain or nervous system disorders. These posters investigate genes and pathways linked to cognitive, behavioral, or neuropsychiatric conditions in humans. Example topics: autism spectrum disorder, Parkinson's disease, tryptophan-kynurenine metabolism, serotonin/dopamine receptor signaling, Rett syndrome, or other neurodevelopmental conditions.

Cancer

Research motivated by understanding cancer biology, specifically oncogenes, tumor suppressor genes, cell cycle regulation, or cancer-related pathways. These posters frame the work around uncontrolled cell growth, tumor development, or cancer risk. Example topics: KRAS oncogene and colon cancer, mismatch repair genes, tumor suppressors (APC, BRCA2), or PI3K pathway in cancer.

Metabolic & Digestive Disease

Research motivated by understanding metabolic or digestive disorders, including conditions affecting the liver, gut, or nutrient processing. These posters focus on pathways such as carbohydrate metabolism, fatty acid oxidation, acid reflux, or nutrient absorption. Example topics: acid reflux disease, fatty acid metabolism disorders, galactosemia, amylase gene expression in diabetes, trehalose catabolism, or liver dysfunction.

Inherited / Genetic Disorder

Research motivated by understanding specific inherited or rare genetic disorders. These posters explicitly name and characterize a clinical genetic condition and use the model organism to understand its mechanisms. Example topics: Zellweger Spectrum Disorder, oculocutaneous albinism (OCA4), or other inborn errors of metabolism.

Other

Research that does not fit into any of the five primary categories. This may include posters focused purely on applied biotechnology method validation, insect physiology without disease framing, or topics that cross multiple categories without a dominant motivation.

Section C: Per-Poster Evidence for Category Assignment

For each poster, a bullet-point list provides specific text evidence supporting the category assignment. For posters with elements of multiple categories, the tie-breaking rationale is noted.

Avetisyan et al. (2023) - CCC SP23

Assigned Category: Neurological & Psychiatric Disease

Note: This poster also showed elements of: Basic Biology (scores: {'Basic Biology': 4}).

- "Expression Analysis of Autism Related Genes Across the D. Melanogaster Midgut"
- "Likewise, ASD-related genes which are primarily expressed in the"

Berbouti et al. (2025) - CCC FA25

Assigned Category: Basic Biology

Note: This poster also showed elements of: Metabolic & Digestive Disease (scores: {'Metabolic & Digestive Disease': 4}).

- "* The online database Flybase provided information about gene orthologs"
- "would show comparable regional expression. We analyzed gene expression"
- "Trehalase Expression in the Drosophila Midgut: Differential Expression"
- "patterns across midgut regions. TREH was highly expressed in the anterior and"
- "posterior midgut, TRET1 was enriched in the copper cell region, and GLUT1 was"
- "posterior midgut, TRET1 was enriched in the copper cell region, and GLUT1 was"
- "patterns across midgut regions. TREH was highly expressed in the anterior and"

Brown et al. (2025) - CCC SP25

Assigned Category: Basic Biology

Note: This poster also showed elements of: Metabolic & Digestive Disease (scores: {'Metabolic & Digestive Disease': 3}).

- "ortholog being ATP6VOC, which encodes a subunit of the vacuolar H'-ATPase (V-ATPase), necessary for"

- “hormonal signaling pathways, vhal6-1 emerges as a core integrator of digestive processes and systemic”
- “V-ATPase, which is a channel protein that encourage multiple functions. Within this we identified paralogs to”
- “processes are closely regulated by gut-brain sensory signaling, which adapts feeding behavior throughout the”
- “melanogaster midgut, it focus on acidification processes that influence metabolic gene regulation. These”

Cabral et al. (2025) - CCC FA25

Assigned Category: Neurological & Psychiatric Disease

Note: This poster also showed elements of: Basic Biology (scores: {‘Basic Biology’: 2}).

- “are linked to Rett syndrome and related neurodevelopmental disorders; however, the”
- “The methyl-CpG binding protein 2 (MECP2) gene is essential for neuronal”
- “beta 3 (Gabrb3) and brain derived neurotrophic factor (Bdnf). Mutstions in MECP2”
- “beta 3 (Gabrb3) and brain derived neurotrophic factor (Bdnf). Mutstions in MECP2”
- “in the same regions, while Bdnf shows higher expression in the prefrontal cortex”
- “including the hippocampus, cerebellum, cortex, striatum, and thalamus. Expression”
- “are linked to Rett syndrome and related neurodevelopmental disorders; however, the”

Camara et al. (2025) - LU SP25

Assigned Category: Basic Biology

Note: This poster also showed elements of: Metabolic & Digestive Disease (scores: {‘Metabolic & Digestive Disease’: 3}).

- “* These are orthologs to human amylases and may relate to diseases”
- “* We hypothesized differential expression of these genes across the”
- “— Amy-d: Linked to gut-brain axis (Leue, 2016).”
- “* Regions compared: anterior (al) vs. copper cell (Cu).”

Dehuelbes et al. (2025) - LU SP5

Assigned Category: Basic Biology

Note: This poster also showed elements of: Metabolic & Digestive Disease (scores: {‘Metabolic & Digestive Disease’: 1}).

- “Regional Expression of Key Metabolic genes in Drosophila Midgut: CRAT, MANF, NPY and their relation to”
- “* NPY Controls appetite, energy balance, circadian”

Diaz et al. (2025) - CCC SP25

Assigned Category: Basic Biology

Note: This poster also showed elements of: Metabolic & Digestive Disease (scores: {‘Metabolic & Digestive Disease’: 1}).

- “examined the regional expression of V-ATPase genes”
- “manages the gut microbiome (3)(6).V-ATPase plays the”
- “highest expression in the Posterior 2-4 (P2_4) midgut region”
- “crucial role of acidification within copper cells; therefore,”
- “highest expression in the Posterior 2-4 (P2_4) midgut region”

Ford et al. (2023) - CCC SP23

Assigned Category: Basic Biology

- "Some Cytochrome P450 genes play an important role in the development"
- "found in al, p1 and p2 4 midgut regions of Drosophila"
- "found in al, p1 and p2 4 midgut regions of Drosophila"

Gill & Alcazar (2025) - CCC FA25

Assigned Category: Neurological & Psychiatric Disease

Note: This poster also showed elements of: Basic Biology (scores: {'Basic Biology': 1}).

- "5-HTR1A receptors which are encoded by HT1A are expressed more in cortical"
- "Differential Expression of Serotonin Receptor Genes Htrla and Htr2a"
- "understand mental health disorders (1). Htr2a on the otherhand isa"
- "to STR showing stronger Serotonergic"
- "Differential Expression of Serotonin Receptor Genes Htrla and Htr2a"
- "in the Prefrontal Cortex and Striatum"
- "in the Prefrontal Cortex and Striatum"
- "Hirla, plays a significant role in mood and anxiety in the brain and"

Godoy-Pena et al. (2025) - CCC FA25

Assigned Category: Basic Biology

Note: This poster also showed elements of: Neurological & Psychiatric Disease, Metabolic & Digestive Disease (scores: {'Neurological & Psychiatric Disease': 1, 'Metabolic & Digestive Disease': 1}).

- "This Neuropeptide in flies"
- "gut-brain region. Finding orthologs to further the evidence of"
- "influence the drosophila fly's circadian rhythm. The genes"
- "Regulation of Sleep Quality: Analysis of Gene Expression Using"
- "gut-brain region. Finding orthologs to further the evidence of"
- "- Confirmed CCHa2-R presence in the copper midgut region"
- "hormone found in the gut/fat of the body. NPF (Neuropeptide F) in flies"
- "NPF for further investigation. We then used Human Protein"

Haubelt & Alcazar et al. (2025) - CCC FA25

Assigned Category: Neurological & Psychiatric Disease

Note: This poster also showed elements of: Basic Biology (scores: {'Basic Biology': 3}).

- "to the neuropsychiatric disorder Parkinson's Disease and wanted"
- "Disease, Schizophrenia, and Post-traumatic Stress Disorder are"
- "Disease, Schizophrenia, and Post-traumatic Stress Disorder are"
- "to the neuropsychiatric disorder Parkinson's Disease and wanted"
- "to the neuropsychiatric disorder Parkinson's Disease and wanted"
- "to the neuropsychiatric disorder Parkinson's Disease and wanted"
- "(Comt), dopamine receptor D2 (Drd2), and dopa decarboxylase"
- "expressed in the striatum part of the mice brain. The striatum is a"

Henriquez et al. (2025) - COD WI24

Assigned Category: Basic Biology

Note: This poster also showed elements of: Metabolic & Digestive Disease (scores: {'Metabolic & Digestive Disease': 1}).

- “6 Immune Pathways Exhibit Similar Patterns of Gene Expressions”
- “suggesting a need for additional immune response in these”
- “The different cell types that reside within the midgut may play”
- “immune pathways to regulate the gut microbiome.”
- “midgut regions of Drosophila melanogaster characterized by”
- “midgut regions of Drosophila melanogaster characterized by”

Holmes (2025) - CCC SP25

Assigned Category: Neurological & Psychiatric Disease

Note: This poster also showed elements of: Basic Biology (scores: {‘Basic Biology’: 8}).

- “disorders such as Parkinson’s. (Moreno-Garcia 2021) This could explain the”
- “Neuromelanin and Dopamine”

Lemus et al. (2025) - FA25 CCC

Assigned Category: Metabolic & Digestive Disease

Note: This poster also showed elements of: Cancer, Basic Biology (scores: {‘Cancer’: 1, ‘Basic Biology’: 6}).

- “the regional expression patterns of two genes both involved in starch and”
- “sucrose metabolism using single-cell as well as bulk RNA-sequencing”
- “the amylase family would share a similar function given their lineage. Upon”
- “digestion as well as metabolic disorders. Our researched aimed to compare”
- “starch hydrolysis, the breakdown of starch into glucose and maltose through the addition of”
- “with the hydrolysis of dietary starch, is significantly differentially expressed in”

Logan et al. (2025) - CCC FA25

Assigned Category: Basic Biology

Note: This poster also showed elements of: Cancer (scores: {‘Cancer’: 1}).

- “development. Since the ACE gene 1s a regulator of fluidity and”
- “fly ortholog of the human ACE gene. We analyzed four ANCE paralogs”
- “- Utilized SciServer, under Differential Expression with DESeq2, to find if the”
- “fly ortholog of the human ACE gene. We analyzed four ANCE paralogs”
- “- To gather more supporting information, we looked at the different cell types”
- “expression patterns. Across different midgut regions, we observed distinct”
- “posterior midgut and epithelial enterocytes. These findings”
- “expression patterns. Across different midgut regions, we observed distinct”

Luera et al. (2025) - CCC FA25

Assigned Category: Basic Biology

- “We analyzed Neuropeptide F (NPF) and its receptor NPFR mm Drosophila”
- “Gene Identification: Using FlyBase, orthologs of mammalian NPY and its”
- “Differential expression analysis revealed that NPF and NPFR were among the most sigmficantly regulated genes across mudzut regions (adjusted p-value $<$ 0.03,”
- “differentially expressed genes between mudzut regions or cell types. Genes were”
- “leaming/memory, and circadian regulation.”
- “and sleep behaviors m Drosophila melanogaster and 1s analogous to mammalian”
- “may serve as fimctional hubs for feeding regulation. Unexpectedly, some midgut regions”
- “We analyzed Neuropeptide F (NPF) and its receptor NPFR mm Drosophila”

Meraz et al. (2023) - CCC SP23

Assigned Category: Cancer

Note: This poster also showed elements of: Metabolic & Digestive Disease, Basic Biology (scores: {'Metabolic & Digestive Disease': 1, 'Basic Biology': 5}).

- "tumor-suppression and is localized to the small"
- "gene that functions as a tumor suppressor (such as FRK/SRC) would"

Nii et al. (2025) - CCC FA25

Assigned Category: Basic Biology

Note: This poster also showed elements of: Metabolic & Digestive Disease (scores: {'Metabolic & Digestive Disease': 2}).

- "functional annotations of mammalian orthologs, allowing comparison of iron"
- "and ATPCL across eight midgut regions (31, 32.3, Cu, LFC, Fe, 0, p24, R4_R5|."
- "and ATPCL across eight midgut regions (31, 32.3, Cu, LFC, Fe, 0, p24, R4_R5|."

Otala et al. (2025) - LU SP25

Assigned Category: Basic Biology

Note: This poster also showed elements of: Cancer, Metabolic & Digestive Disease, Inherited / Genetic Disorder (scores: {'Cancer': 1, 'Metabolic & Digestive Disease': 6, 'Inherited / Genetic Disorder': 1}).

- "mechanisms and supports the development of therapeutic"
- "data for human orthologs [Chintapalli, et al. 2007]"
- "Fig. 1 ClustrProfiler results for differential expression analysis between a1 (anterior"
- "Fig 2 Acox3 expression across the Drosophila midgut regions. Acox3 expression"
- "» Compared gene expression between al (anterior midgut)"
- "and p1 (posterior midgut) regions"
- "Fig 2 Acox3 expression across the Drosophila midgut regions. Acox3 expression"

Paderna et al. (2025) - CCC SP25

Assigned Category: Inherited / Genetic Disorder

Note: This poster also showed elements of: Basic Biology (scores: {'Basic Biology': 5}).

- "with oculocutaneous albinism type 4, a"
- "(OCA4) manifests as reduced pigment in the skin,"
- "Albinism and Melanogenesis Pathway:"

Paramo-Ojeda et al. (2025) - CCC SP25

Assigned Category: Cancer

Note: This poster also showed elements of: Metabolic & Digestive Disease, Basic Biology (scores: {'Metabolic & Digestive Disease': 1, 'Basic Biology': 6}).

- "~~ Model for KRAS-Driven Colon Cancer"
- "lead to tumor development and are often associated with poor"
- "most frequently mutated oncogenes in human cancers is"
- "~~ Model for KRAS-Driven Colon Cancer"
- "Ras85D Expression in Drosophila Midgut Highlights a"
- "~~ Model for KRAS-Driven Colon Cancer"
- "including BRCA2, APC, Pi3K92E (PIK3CA), MLH]1, and PMS2,"
- "including BRCA2, APC, Pi3K92E (PIK3CA), MLH]1, and PMS2,"

Pedireddi et al. (2025) - CCC SP25

Assigned Category: Metabolic & Digestive Disease

Note: This poster also showed elements of: Basic Biology (scores: {'Basic Biology': 2}).

- "15 genes In the starch ond sucrose metabolic pathway. We found 7 genes"
- "15 genes In the starch ond sucrose metabolic pathway. We found 7 genes"
- "Beyond the Anterior: Amylase Gene Expression Suggests Widespread"
- "Polysaccharide Digestion in the Drosophila Midgut |"
- "the P region was also responsible for nutrient absorption (Marianes,"

Rodriguez (2025) - CCC SP25

Assigned Category: Metabolic & Digestive Disease

Note: This poster also showed elements of: Inherited / Genetic Disorder, Basic Biology (scores: {'Inherited / Genetic Disorder': 1, 'Basic Biology': 1}).

- "Galactosemia is a rare metabolic disorder that results from deficiencies in the"
- "the disease galactosemia, which prevents the body from properly metabolising"
- "glucose-1-phosphate."

Sakana et al. (2025) - CCC SP25

Assigned Category: Basic Biology

- "Gene expression patterns that control development and"
- "melanogaster known for its role in tissue patterning, stem cell"
- "Differential Expression of the TALE Homeobox Genes in"
- "Integration Site genes (MEIS] and MEIS2), orthologs of"
- "Differential Expression of the TALE Homeobox Genes in"
- "differentiation are often orchestrated by transcription factors."
- "gene paralogs in Drosophila. Drosophila melanogaster serves as"
- "Extension (TALE) homeobox genes across the midgut regions."

Trevino et al. (2022) - CCC SP22

Assigned Category: Inherited / Genetic Disorder

Note: This poster also showed elements of: Metabolic & Digestive Disease, Basic Biology (scores: {'Metabolic & Digestive Disease': 3, 'Basic Biology': 7}).

- "'Drosophila Melanogaster a Good Model System of Zellweger Spectrum Disorder"
- "ABCD1, ACBDS, ACOX1, DNM1L, HSD17B4, and SCP2 are listed as peroxisomal disorders"
- "ZSD is a rare inherited disorder characterized by the absence/reduction of functional"
- "Zellweger Spectrum Disorder (ZSD) is a genetic disorder that is caused by the mutation"
- "of 1 out of 13 different genes involved in the formation and function of peroxisomes."
- "ABCD1, ACBDS, ACOX1, DNM1L, HSD17B4, and SCP2 are listed as peroxisomal disorders"

Tuttle et al. (2025) - CCC SP25

Assigned Category: Neurological & Psychiatric Disease

Note: This poster also showed elements of: Metabolic & Digestive Disease, Basic Biology (scores: {'Metabolic & Digestive Disease': 1, 'Basic Biology': 5}).

- "Gene Expression Patterns in the Tryptophan-Kynurenine"
- "Gene Expression Patterns in the Tryptophan-Kynurenine"

Zlaket et al. (2023) - CCC SP23

Assigned Category: Inherited / Genetic Disorder

Note: This poster also showed elements of: Metabolic & Digestive Disease, Basic Biology (scores: {'Metabolic & Digestive Disease': 2, 'Basic Biology': 8}).

- "can cause Zellweger's Syndrome, a buildup in several fatty acids like"
- "in | oxidation of fatty acid chains within peroxisomes, and SCP2, a sterol and"
- "ScpX: A Peroxisomal Gene and its Distinctive Paralogs"